Practical 4

Time Series Features: Spectral Analysis

Getting started

At the end of this practical you should be able to:

1. Interpret the spectrum plot (smoothed periodogram) for real (transformed) data.
2. Interpret the spectrum plot (smoothed periodogram) for model residuals.
3. Utilise unit root tests to identify if differencing is required.
4. Use plots and models to investigate structure in time series data.

Before getting started, you may find it beneficial to read these some documentation of the packages we will use:

- ggtime website
  - `ggtime` 0.1.0 (O'Hara-Wild et al. 2025)[1]

The Mauna Loa Data

Recall for the Mauna Loa Data, and some of the models developed in the previous practical.

```
library(tsibble)
library(dplyr)
library(fable)
library(ggplot2)
library(ggtime)

mauna <- as_tsibble(co2) |>
  # use meaningful names
  rename(year_month = index, concentration = value)

mauna_models <- mauna |>
  model(
    lm_cs = TSLM(concentration ~ trend() + I(trend()^2) + I(trend()^3) + season()),
    arima1 = ARIMA(concentration ~ pdq(1,1,0) + PDQ(0,1,0))
    )
```

1. Obtain time series plots, ACFs, and spectrum plots using

- `gg_tsdisplay(plot_type = "spectrum")` for the original, differenced and seasonally differences data, and
- `select(model_name) |> gg_tsresiduals(plot_type = "spectrum")` for the residual of the models.

2. Conduct unit root tests to identify the need for (further, seasonal) differencing.

3. Compare the spectrum plots and comment on what additional structure is remaining in the model apply after transformations or modelling.

Analysis of deaths due to lung disease

The data on monthly deaths due to lung disease in the UK is available as the `ts` object, `ldeaths`.

4. Convert `ldeaths` to a `tsibble`.

5. Obtain a time series plot of the data and comment on the salient features of the plot.

6. Obtain an ACF plot (correlogram) and a spectrum plot (smoothed periodogram). Of these plots, which do you consider to be more informative for these data? Explain.

7. What steps would you take next with this data? Apply an analysis of your own choosing, checking its viability through the use of residual, ACF, and spectrum plots and tests as appropriate.

8. Are the any interesting aspects of the time series to report? Do you think it is possible to forecast the time series with the model you have chosen?

References

O'Hara-Wild, Mitchell, Cynthia A. Huang, Matthew Kay, and Rob Hyndman. 2025. *Ggtime: Grammar of Graphics and Plot Helpers for Time Series Visualization*. https://doi.org/10.32614/CRAN.package.ggtime .

Footnotes

1. In particular the documentation for `ggtime::gg_tsdisplay` and `ggtime::gg_tsresiduals`. ↩